ShopNest Business Analytics Dashboard
Power BI Capstone Project Report

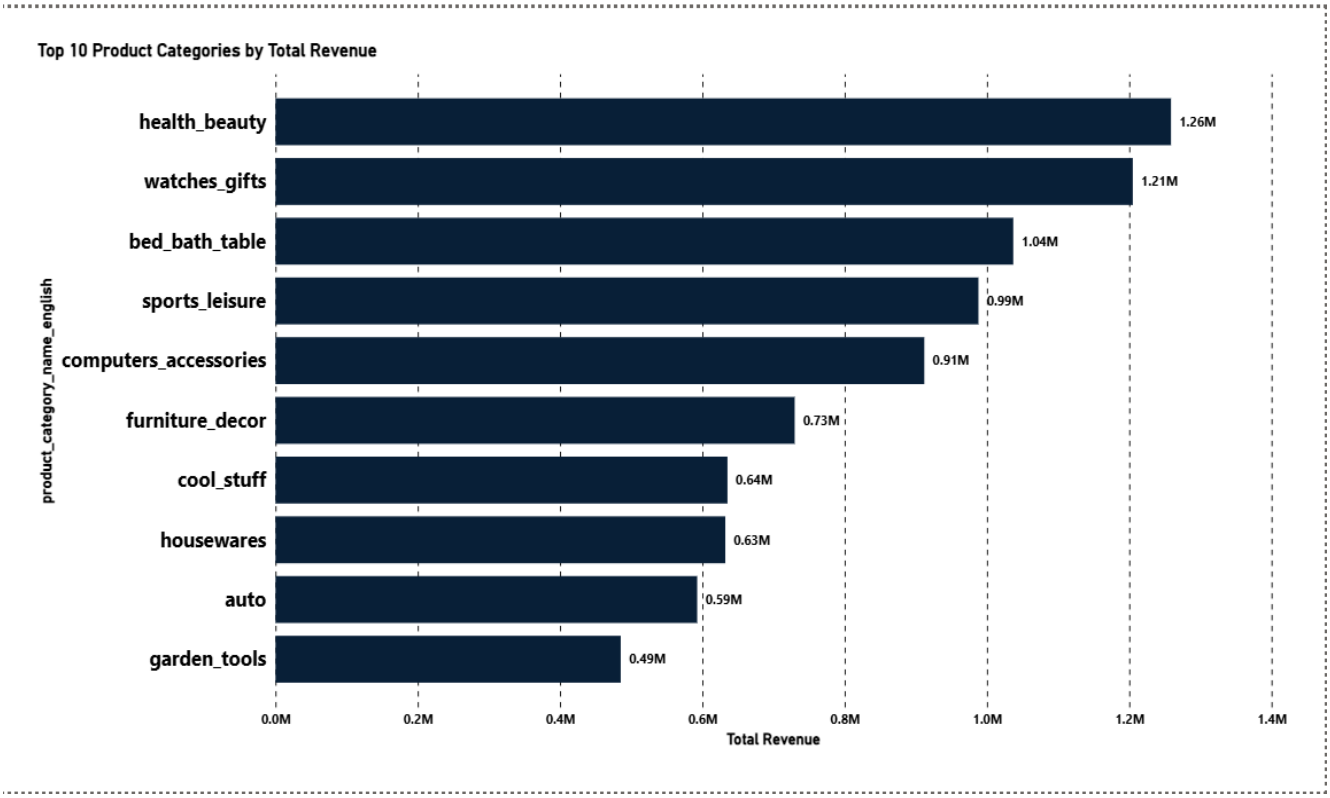
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Batch No. : DA-202604-1
Tools used : Microsoft Power BI
Dataset : ShopNest E-Commerce Dataset
Duration : 29th June 2026 – 13th July 2026

Introduction

ShopNest is one of the leading e-commerce marketplaces in Portugal that connects customers with sellers across different regions. This Power BI dashboard was developed to analyse the ShopNest e-commerce dataset and provide meaningful business insights. The dashboard answers key analytical questions related to revenue, product categories, delayed orders, payment methods, product ratings, state-wise sales, seasonal sales patterns, and yearly revenue trends. Interactive slicers for Year, Category, and State allow users to explore the data dynamically.

Question 1: Top Categories by Total Revenue.

Question Statement: Identify and visually represent the Top 10 Product Categories by Total Revenue.

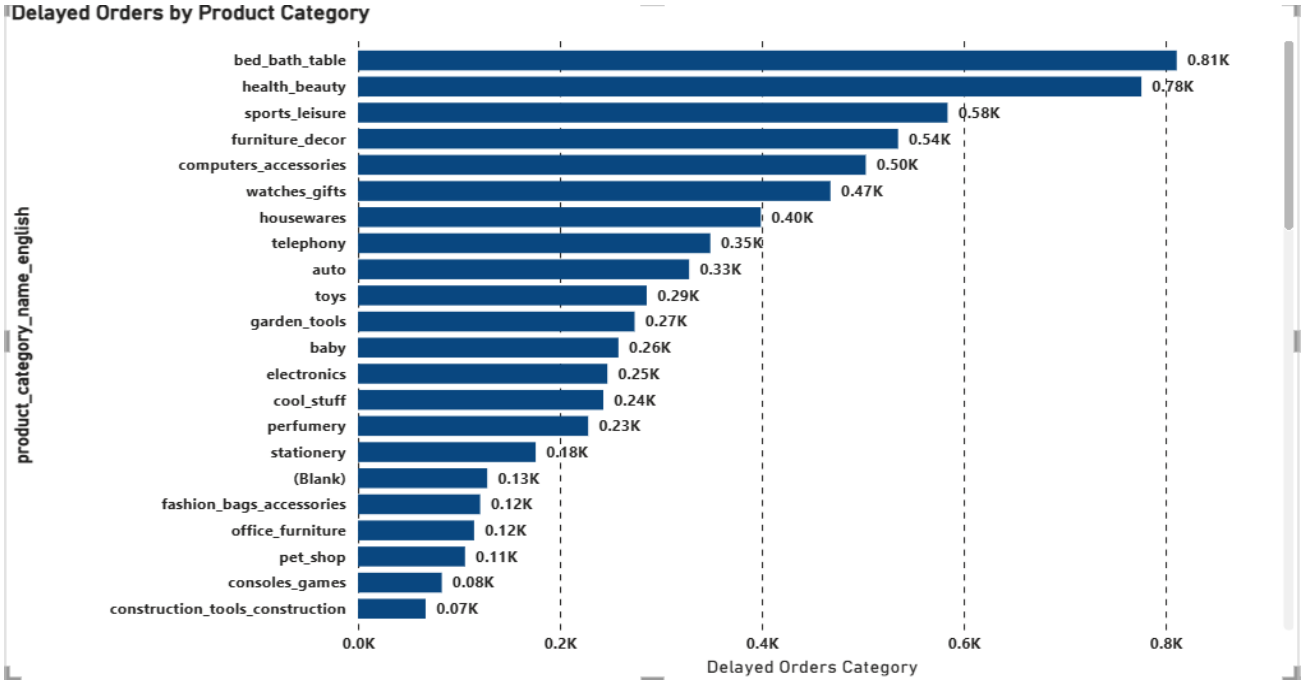


Explanation

This visual displays the Top 10 Product Categories based on Total Revenue generated by ShopNest. The **Health Beauty** category generated the highest revenue (approximately **1.26 million**), followed by **Watches Gifts** (approximately **1.21 million**) and **Bed Bath Table** (approximately **1.04 million**). The chart clearly highlights the best-performing product categories, enabling the business to identify high-demand products, prioritize inventory, and focus marketing efforts on the most profitable categories.

Question 2: **Delayed Orders by Product Category.**

Question Statement: **Determine the number of delayed orders in each product category. An order is considered delayed if the actual delivery date is later than the estimated delivery date.**

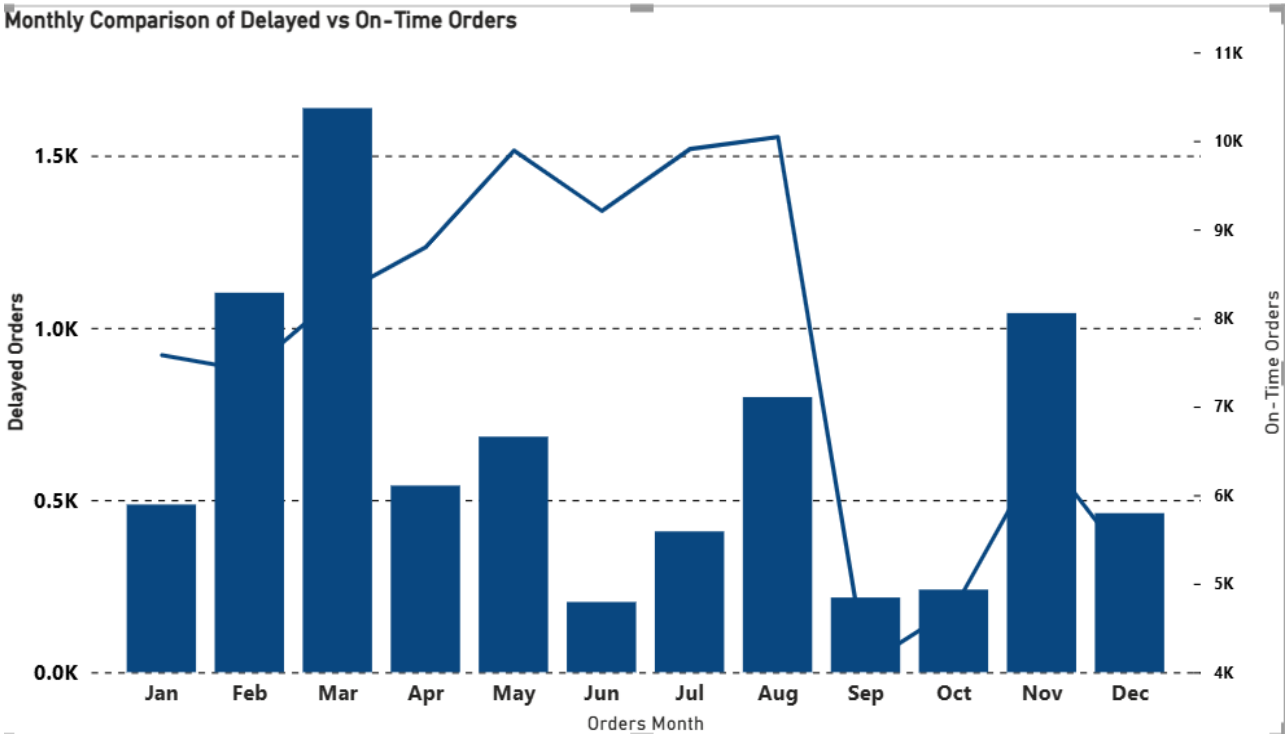


Explanation

This visual displays the number of delayed orders in each product category. An order is considered delayed when the actual delivery date is later than the estimated delivery date. The Bed Bath Table category recorded the highest number of delayed orders (approximately 810), followed by Health Beauty (approximately 780) and Sports Leisure (approximately 580). Other product categories also experienced delivery delays but at comparatively lower levels. This analysis helps the business identify categories with frequent delivery issues, enabling improvements in logistics, inventory management, and overall customer satisfaction.

Question 3: Monthly Comparison of Delayed vs On-Time Orders.

Question Statement: Create a dynamic visual that compares the number of delayed orders and on-time orders for each month. Utilize the drill-through cross-report feature to provide a detailed analysis of late and on-time deliveries.

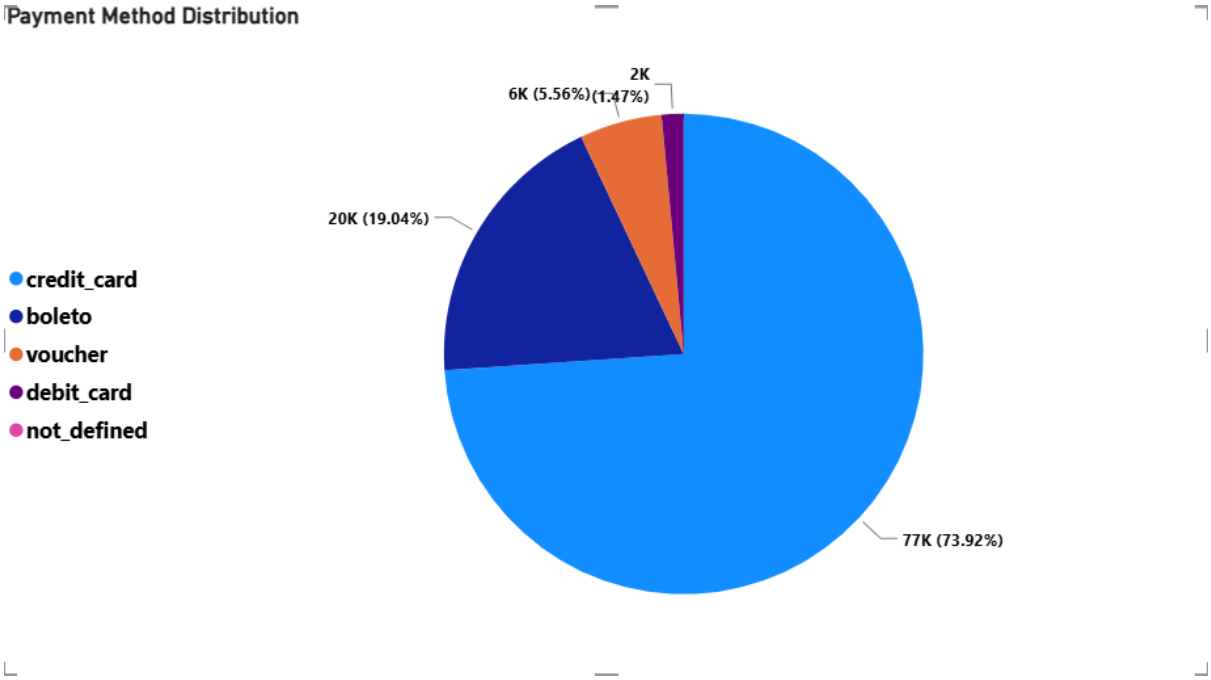


Explanation

This visual compares the monthly number of delayed and on-time orders. Using the drill-through cross-report feature, users can further analyse late and on-time deliveries for each month. The chart shows that delayed orders peaked in March, while September and October recorded the fewest delayed orders. The on-time delivery trend also varies across the months, indicating seasonal changes in delivery performance. This analysis helps the business identify periods with higher delivery delays, improve logistics planning, and maintain consistent customer service throughout the year.

Question 4: Payment Method Analysis.

Question Statement: Analyse the most frequently used payment methods by customers using a visually appealing representation such as a pie chart or other suitable visual.

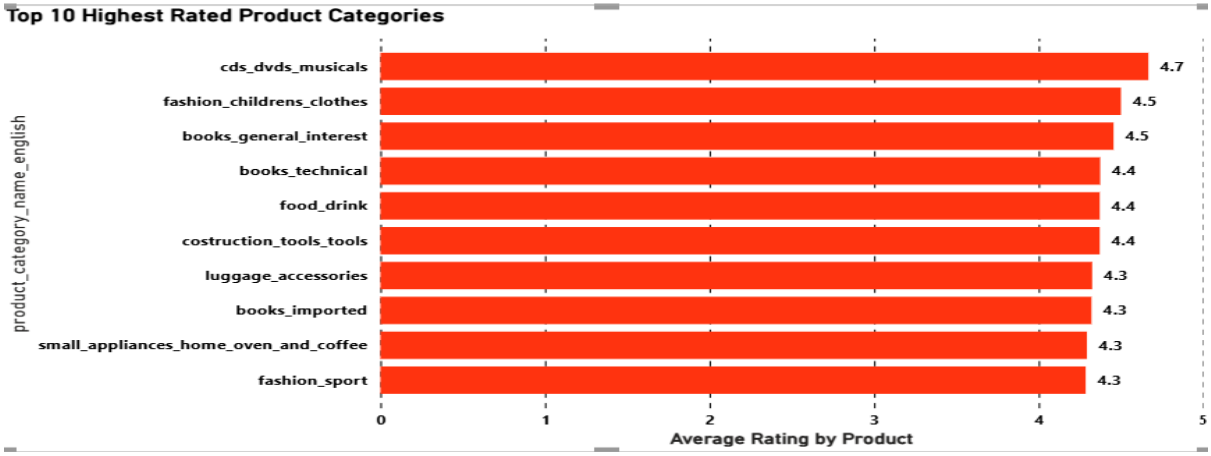


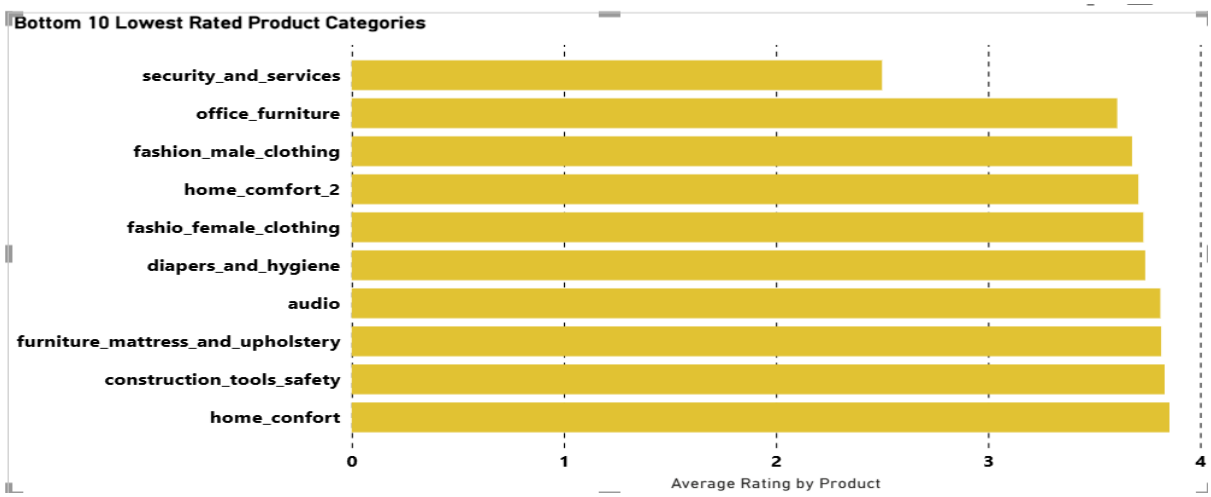
Explanation

This visual illustrates the most frequently used payment methods by customers for purchases on ShopNest. The **Credit Card** payment method is the most frequently used, accounting for approximately **74%** of all transactions, followed by **Boleto** with approximately **19%** and **Voucher** with approximately **5%**. **Debit Card** and **Not Defined** payment methods contribute only a very small percentage of the total transactions. This analysis helps the business understand customer payment preferences, optimize payment services, and focus on supporting the payment methods most preferred by customers, thereby improving the overall customer purchasing experience.

Question 5: Product Ratings Analysis.

Question Statement: Determine the Top 10 Highest Rated Product Categories and the Bottom 10 Lowest Rated Product Categories using bar charts.



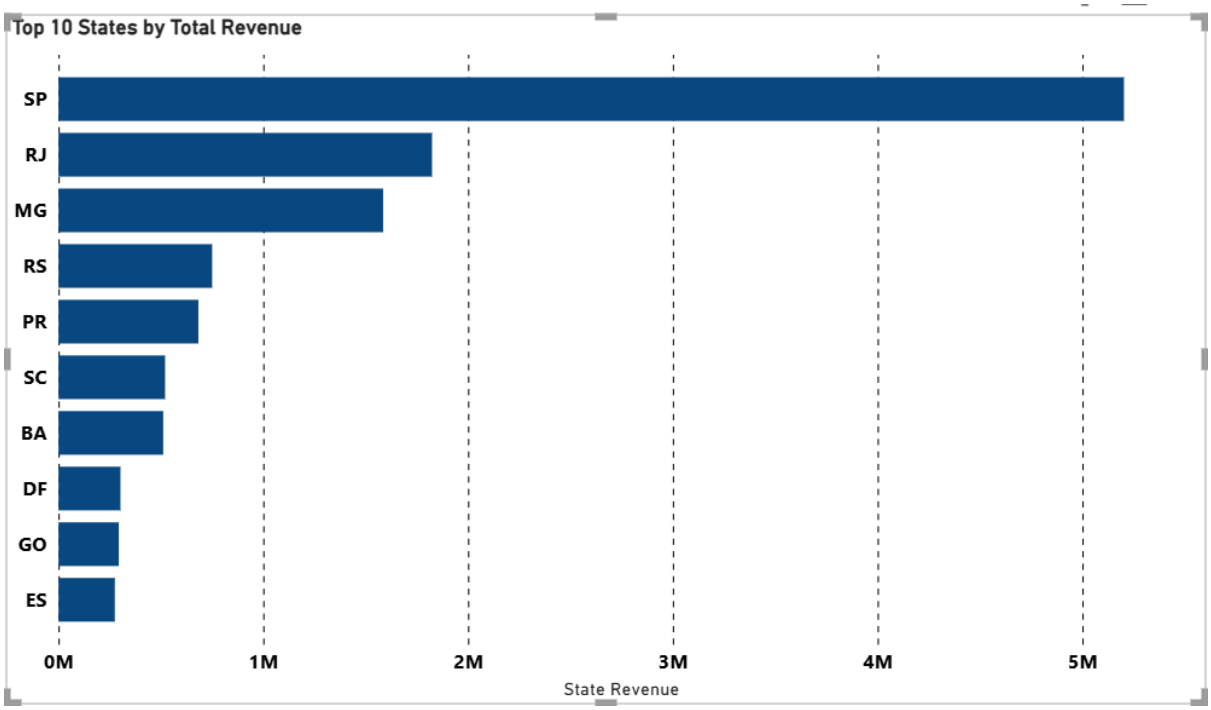


Explanation

These visuals compare the **Top 10 Highest Rated** and **Bottom 10 Lowest Rated Product Categories** based on average customer ratings. Categories such as **CDs/DVDs/Music**, **Fashion Children's Clothing**, and **Books** received the highest average ratings, indicating strong customer satisfaction and positive product quality. In contrast, categories such as **Security and Services**, **Office Furniture**, and **Fashion Male Clothing** received comparatively lower ratings, suggesting areas where customer experience may require improvement. This analysis helps the business identify its best-performing product categories, recognize products that need attention, and make informed decisions to improve product quality, customer satisfaction, and overall business performance.

Question 6: State-wise Sales Analysis.

Question Statement: Identify and visually represent the states with the highest and lowest sales to provide a clear understanding of regional sales performance.

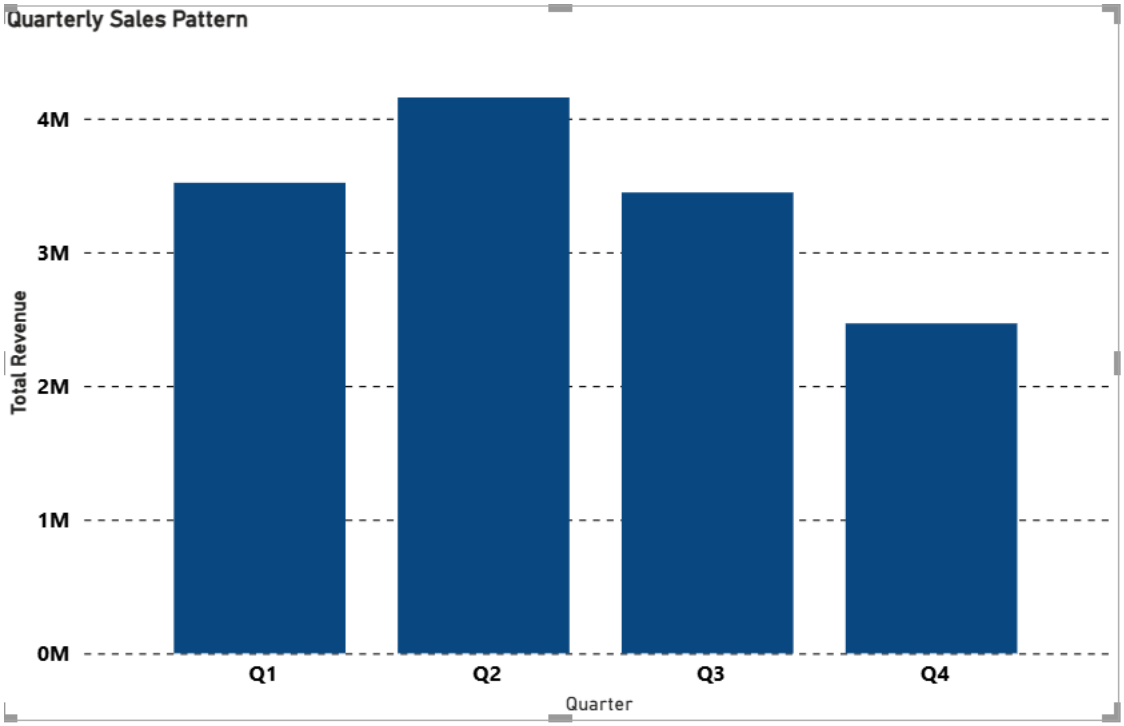


Explanation

This visual displays the **Top 10 States by Total Revenue**, highlighting the regional distribution of sales across ShopNest. **São Paulo (SP)** generated the highest revenue by a significant margin, followed by **Rio de Janeiro (RJ)** and **Minas Gerais (MG)**. The remaining states contributed comparatively lower sales, indicating variations in regional market performance. This analysis helps the business identify high-performing regions, allocate resources effectively, develop targeted marketing strategies, and explore growth opportunities in lower-performing states to improve overall sales performance.

Question 7: Seasonal Sales Pattern.

Question Statement: Investigate and visually represent the quarterly sales pattern to identify seasonal trends in sales performance throughout the year.

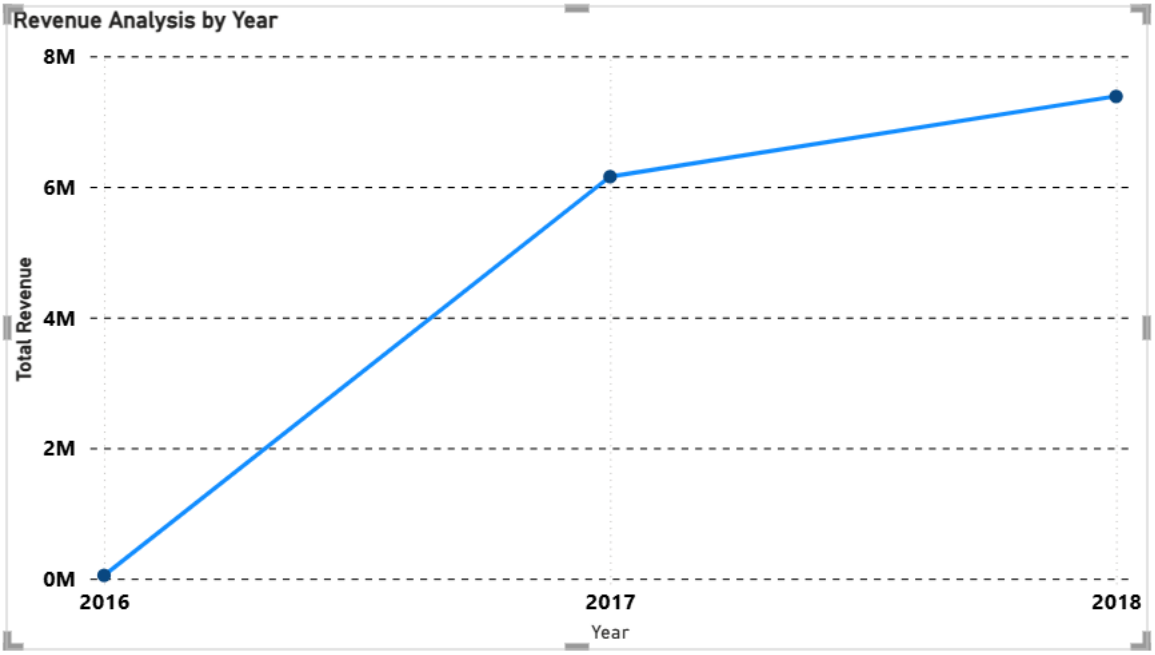


Explanation

This visual illustrates the **quarterly sales pattern** throughout the year, helping identify seasonal trends in sales performance. **Quarter 2 (Q2)** recorded the highest total sales (approximately **4.1 million**), followed by **Q1** (approximately **3.5 million**) and **Q3** (approximately **3.4 million**). **Quarter 4 (Q4)** generated the lowest sales (approximately **2.4 million**), indicating a noticeable decline in revenue during the final quarter. This analysis helps the business understand seasonal demand patterns, improve sales forecasting, optimize inventory planning, and develop targeted marketing strategies to maximize revenue during high-performing periods and strengthen performance during weaker quarters.

Question 8: Revenue Analysis.

Question Statement: Determine the total revenue generated by ShopNest and analyse how it changed over time (yearly). Represent this information through suitable visuals to highlight trends and patterns.



Explanation

This visual illustrates the yearly revenue trend generated by ShopNest. Revenue increased significantly from 2016 to 2017 and continued to grow in 2018, indicating consistent business growth over the three-year period. The highest revenue was recorded in 2018 (approximately 7.6 million), followed by 2017 (approximately 6.2 million), while 2016 recorded the lowest revenue due to fewer transactions during the initial period. This analysis helps the business evaluate long-term revenue growth, measure overall business performance, identify revenue trends over time, and support strategic planning for future expansion and sales forecasting

Conclusion

This report analysed ShopNest's sales performance using Power BI visualizations. The analysis identified the top-performing product categories, delayed orders, payment method preferences, product ratings, regional sales performance, seasonal sales patterns, and yearly revenue trends. These insights provide valuable information for improving business decisions, enhancing customer satisfaction, optimizing inventory and logistics, and supporting future sales and marketing strategies.

End of Report